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TripleNet: A Low-Parameter Network for Low Computing Power Platform

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***Abstract*—Deep learning has achieved great success in computer vision (CV), and convolutional neural networks (CNN) have become an important network architecture for CV tasks. With the widespread use of mobile devices, neural network models based on low-computing power platforms have received much attention. This paper proposes a lightweight CNN model named TripleNet. TripleNet learns the model compression methods of HarDNet and ThreshNet and improves on them. TripleNet completes the network architecture by combining three different convolutional layers, which has the advantages of small model size and high test accuracy. With similar test accuracy, TripleNet model has fewer parameters than HarDNet and ThreshNet. This paper conducts image classification experiments using TripleNet and other networks on CIFAR-10 and SVHN datasets. The experimental results show that compared with HarDNet, the parameters of TripleNet are reduced by 66% and the accuracy rate is improved by 18%; compared with ThreshNet, the parameters of TripleNet are reduced by 37% and the accuracy rate is improved by 5%.**

Keywords—convolution neural network, image classification, low computing power platforms, computer vision, deep learning

I. Introduction

Mobile applications usually rely directly on cloud servers to complete all computing operations of deep learning. This method does not require a mobile device with powerful computing power to complete the deep learning calculation, but the disadvantage is that the data transmission cost between the mobile device and the cloud server is relatively high. Therefore, we noticed that it is very important to design neural networks suitable for low computing power platforms, using the computing power of mobile devices rather than relying on cloud services.

The research of deep learning (DL) in computer vision has gradually matured, but most of works only pursues the accuracy of neural networks to complete CV tasks. Therefore, the network architecture is designed to be large, and the parameters of the model are usually ignored.

As one of the state-of-the-art CNN, DenseNet [1] has the characteristics of promoting feature reuse. It has achieved excellent performance in image classification and object detection tasks. However, the interconnection of each layer in its architecture makes the network require large memory hardware. And the training time and inference time of the DenseNet model are too long, making it impossible to use in practical applications. To reduce the inference time per image of the model, HarDNet [2] reduces the connections between layers, and resets the weights of layers to increase feature extraction to maintain model accuracy. ThreshNet [3] utilizes Threshold Mechanism to determine the connections between layers and resets the number of channels. However, both of these two neural networks ignore the problem of excessive parameters. To solve this problem, we design different convolutional layers to form different blocks, improving model accuracy while reducing the amount of model parameters.

II. TripleNet

A. Architecture

TripleNet model architecture consists of 5 blocks. We propose three different convolutional layers to compose blocks, the composition method is shown in Figure 1. Ablation experiments show that the convolutional layer composition method of the first block is the most effective to improve the accuracy, but it will take up a lot of memory. The convolution composition method of the second, third and fourth blocks occupies less memory. In line with the increase in the number of channels, using this method in these blocks is more suitable for low computing power platforms. The last block uses residual learning, because its input feature map is large, and using a specific combination of convolutional layers can ensure that the neural network has less parameters and less model computation.

The CNN architecture obtains a smaller size feature map by downsampling and extracts the feature map for calculation. TripleNet plays the role of downsampling through a 1 × 1 convolutional layer and a 2 × 2 average pooling layer before the block.

Table 1. Details of TripleNet model architecture

Model	Layer	Channel	Growth Rate
TripleNet-S	6, 16, 16, 16, 2	128, 192, 256, 320, 720	32, 16, 20, 40, 160
TripleNet-B	6, 16, 16, 16, 3	128, 192, 256, 320, 1080	32, 16, 20, 40, 160

B. Convolution Layer

For the first block, we use a 3 × 3 convolutional layer first, whose output is *Growth Rate*, the specific value is shown in Table 1. After this, we add a 1 × 1 convolutional layer with a fixed output of 4 × *Growth rate*. We add BN and ReLU before these two convolutional layers to generate better feature selection. The first block uses a dense connection approach, which is that the input to the next convolutional layer is the feature maps of all previous layers.

In order to reduce the model size and shorten the inference time per image, the convolutional layer in the middle three blocks adopts the architecture of Convolution-BN-ReLU, and only a single 3 × 3 convolution is used. These three blocks adopt the method of harmonic dense connection, and the fixed output of the convolution that retains the input connection is 1.7 × *Growth rate*.

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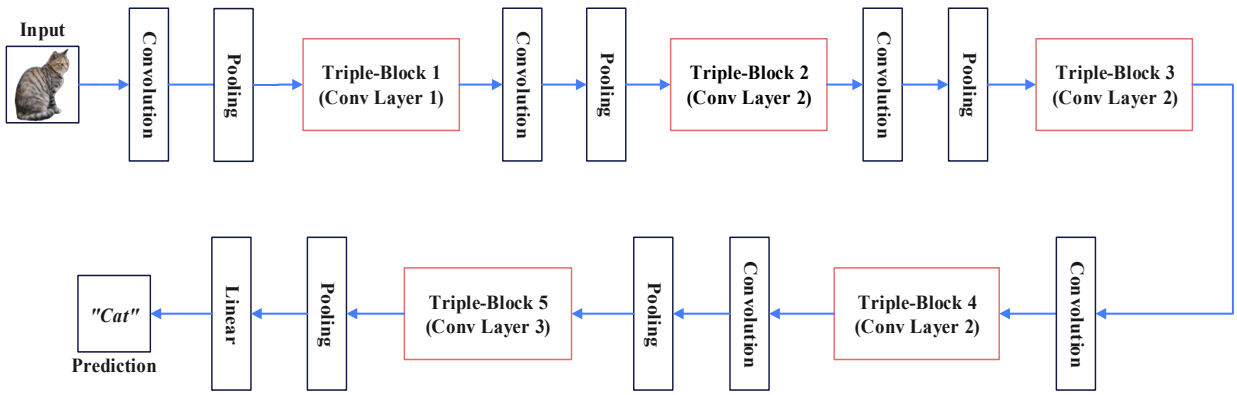

Figure 1. The overall architecture of TripleNet.

In order to reduce the amount of parameters of the model and the amount of model calculation, the depth of the last block is only set to 2 or 3. To ensure the accuracy of the model, the convolutional layer uses residual learning. Since 1 × 1 convolutions are computationally less computationally intensive, this architecture does not generate a large number of parameters.

Table 2. Comparison of parameters and experimental results

Model	Params	CIFAR Error	CIFAR Inference Time	SVHN Error	SVHN Inference Time
TripleNet-S	9.67M	**13.05%**	**22ms**	**6.18%**	**31ms**
ThreshNet-79	15.32M	13.67%	21ms	6.41%	28ms
HarDNet-85	17.57M	14.65%	27ms	6.97%	27ms
TripleNet-B	12.63M	**12.97%**	**24ms**	**5.67%**	**32ms**
ThreshNet-95	17.14M	13.31%	24ms	6.07%	29ms
ResNeXt-50	25.02M	16.96%	23ms	8.28%	34ms
HarDNet-85	36.67M	13.89%	29ms	6.91%	28ms
Wide_ResNet-50	68.88M	17.31%	25ms	8.31%	28ms

Our network results are in **boldface**, surpassing all competing methods in blue.

III. Experiment

A. Experiment Setup

CIFAR-10 is a dataset consisting of color images of size 32 × 32, with 50,000 images for training and 10,000 images for testing, with a total of 10 classes. The Street View House Numbers (SVHN) is also a dataset consisting of color images of size 32 × 32, with 73,257 training images and 26,032 testing images. For fair comparison, all models are not pre-trained. On the CIFAR-10 dataset, we use a batch of 100, an initial learning rate of 1×10^{-3}, and reduce it by 0.1 every 75 epochs, Adam to optimize the model. All models run on a single GPU RTX 3080Ti for 200 epochs. On the SVHN dataset, we use a batch size of 256, other hyperparameters are the same as above, and use a single GPU RTX 3080Ti for 60 epochs.

B. Experiment Results

In Table 2, we show a comparison of CIFAR-10 and SVHN results for network models ordered by the number of parameters, including TripleNet-S and TripleNet-B model. TripleNet is more accurate than both ThreshNet and HarDNet due to the use of three different convolutional layer combinations instead of a single kind of convolutional layer architecture. On CIFAR-10 dataset, TripleNet-S is 5% more accurate than ThreshNet-79 and 11% more accurate than HarDNet. On SVHN dataset, TripleNet-B is 7% more accurate than ThreshNet-95 and 18% more accurate than HardNet-85.

In the experimental results of image classification, although the inference time is the same as ThreshNet and HarDDNet, the number of parameters is greatly reduced. Compared with HarDNet-85, TripleNet-B reduces the number of parameters by 66%. This experimental result is in line with the original intention of our model design, which is to achieve efficient performance with limited computation.

IV. Conclusion and Future Work

As a backbone network, TripleNet can be applied in different scenarios, such as the Mask R-CNN algorithm architecture to complete instance segmentation, the YOLOv3 algorithm architecture to complete object detection, the Panoptic FPN algorithm architecture to complete the panoptic segmentation. Unlike HarDNet and ThreshNet, whose model parameters are too large, TripleNet has a smaller model size and can run on platforms with low computing power, such as mobile phones, Raspberry Pi, and FPGA. TripleNet is a network architecture designed to obtain efficient performance with limited memory and computing power, and is more suitable for real-life scenarios.

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